
PHAROS: Navigated Hierarchical Attention with Adaptive Budgets, Cross-Level Guidance, Tidal Momentum, and Uncertainty-Aware Fallback

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Abstract

1 Lighthouse Attention introduced symmetric hierarchical pooling with gradient-free
2 top- k selection, enabling efficient long-context pretraining and faster attention com-
3 putation at extended context scale. Despite these gains, three structural rigidities
4 limit its potential: (i) a uniform per-layer top- k budget that ignores head-to-head
5 entropy variance; (ii) independent pyramid-level scoring that discards coarse selec-
6 tion priors; and (iii) no mechanism to propagate selection state across transformer
7 depth, despite empirical evidence that the same tokens often recur across layers.
8 We present PHAROS, a composable augmentation of Lighthouse Attention that
9 addresses these limitations through four coordinated mechanisms: Depth-Sounding
10 Budget Allocation, Range Light Cross-Level Guidance, Tidal Momentum Selec-
11 tion, and Fog Gate Confidence Fallback. Together, these mechanisms establish an
12 adaptive, guided, persistent, and uncertainty-aware hierarchical attention frame-
13 work for improving long-context selection quality while preserving compatibility
14 with efficient transformer attention pipelines.

15 1 Introduction

16 Scaling language model context from tens of thousands to million-token ranges is primarily bottle-
17 necked by the $\Theta(N^2)$ compute and memory of scaled dot-product attention (SDPA) [Vaswani et al.,
18 2017, Dao et al., 2022]. Lighthouse Attention [Peng et al., 2026] introduced a compelling alternative
19 for the pretraining phase: pool queries, keys, and values symmetrically across an L -level pyramid;
20 score the pooled representations with a parameter-free ℓ_2 -norm or dilated scorer; gather the top- k
21 pooled tokens; and run standard FlashAttention on the resulting dense subsequence. After pretraining,
22 a dense-SDPA resume phase recovers a fully dense inference-ready checkpoint.

23 Lighthouse Attention demonstrates improved long-context training efficiency under hierarchical
24 selection [Peng et al., 2026]. However, three design rigidities remain.

25 **Uniform k across heads.** The top- k budget is identical for every head in every layer. Yet attention
26 heads are empirically heterogeneous in focus width [Voita et al., 2019, Clark et al., 2019]: some
27 heads attend to single anchor tokens, while others distribute attention broadly. A single global k can
28 be wasteful for focused heads and insufficient for diffuse heads.

29 **Independent pyramid levels.** Coarse-level pooled representations aggregate larger context regions
30 and therefore encode global selection priors. In Lighthouse Attention, however, coarse-level selection
31 signals are not explicitly used to guide fine-level routing.

32 **Stateless selection across depth.** Selection is computed independently in each transformer layer.
 33 Prior work suggests that important tokens often recur across consecutive layers [Geva et al., 2022,
 34 Yu et al., 2023]. A system that accumulates cross-layer selection statistics may warm up faster and
 35 generalize better.

36 We present PHAROS, short for **P**ooled **H**ierarchical **A**ttention with **R**outed, **O**rche**S**trated **S**election.
 37 PHAROS is a composable augmentation of Lighthouse Attention addressing all three limitations
 38 while adding uncertainty-awareness.

39 Our contributions are:

- 40 • **Depth-Sounding Budget Allocation (DBA):** an entropy-aware per-head k assignment
 41 mechanism with a global budget constraint.
- 42 • **Range Light Cross-Level Guidance (RCG):** a learned additive bias from coarse to fine
 43 pyramid levels, formalized as KL-regularized logit injection.
- 44 • **Tidal Momentum Selection (TMS):** a layer-to-layer exponential moving average of selec-
 45 tion logits, providing a persistent selection prior.
- 46 • **Fog Gate Confidence Fallback (FCF):** a learned confidence mechanism that dynamically
 47 widens the attention beam under uncertainty.

48 PHAROS is implemented as a thin wrapper over the existing Lighthouse-style selection interface and
 49 requires no changes to the attention kernel or the two-stage training recipe.

50 2 Background: Lighthouse Attention

51 We briefly recapitulate Lighthouse Attention to fix notation. Let N be the sequence length and H the
 52 number of attention heads. For head h , denote queries, keys, and values as

$$Q_h, K_h, V_h \in \mathbb{R}^{N \times d}.$$

53 **Pyramid construction.** Level- ℓ pooled representations are computed by average pooling with
 54 stride p^ℓ along the sequence dimension:

$$\bar{X}_h^{(\ell)} = \text{AvgPool}_{p^\ell}(X_h) \in \mathbb{R}^{N/p^\ell \times d}, \quad X \in \{Q, K, V\}. \quad (1)$$

55 **Scoring.** For each query position t , the scorer computes per-pooled-token logits. With the norm
 56 scorer:

$$s_{h,t,j}^{(\ell)} = \left\| \bar{Q}_{h,t}^{(\ell)} \right\|_2 \cdot \left\| \bar{K}_{h,j}^{(\ell)} \right\|_2, \quad j = 1, \dots, N/p^\ell. \quad (2)$$

57 Scores across all L levels are concatenated to form a joint logit vector

$$s_{h,t} \in \mathbb{R}^P, \quad P = \sum_{\ell} N/p^\ell.$$

58 **Selection and gather.** A fixed top- k selection yields the index set

$$I_{h,t} = \text{top-}k_k(s_{h,t}). \quad (3)$$

59 The gathered subsequence $\hat{K}_{h,t}, \hat{V}_{h,t} \in \mathbb{R}^{k \times d}$ is passed to standard FlashAttention.

60 3 PHAROS: Four Coordinated Mechanisms

61 3.1 Depth-Sounding Budget Allocation

62 **Motivation.** Let

$$p_{h,t} = \text{softmax}(s_{h,t})$$

63 be the selection probability distribution of head h at position t . The Shannon entropy

$$\mathcal{H}_{h,t} = -p_{h,t}^\top \log p_{h,t}$$

64 quantifies how spread out the selection signal is. A low-entropy head has a sharp focus and can attend
 65 accurately with a smaller budget, whereas a high-entropy head is uncertain and may benefit from a
 66 larger subsequence.

67 **Formulation.** Let $k_{\text{total}} = H \cdot k_{\text{base}}$ be the global token budget. For each head h , compute the
68 normalized inverse entropy weight:

$$w_h = \frac{\exp(-\hat{\mathcal{H}}_h / \tau_{\text{dba}})}{\sum_{h'} \exp(-\hat{\mathcal{H}}_{h'} / \tau_{\text{dba}})}, \quad \hat{\mathcal{H}}_h = \mathbb{E}_t[\mathcal{H}_{h,t}]. \quad (4)$$

69 A per-head budget is then assigned as

$$k_h = \max(k_{\min}, \text{round}(k_{\text{total}} \cdot w_h)). \quad (5)$$

70 The term $k_{\min}$ prevents degenerate zero-budget heads. As $\tau_{\text{dba}} \rightarrow \infty$, all heads receive approximately
71 uniform budget, recovering the Lighthouse-style allocation.

72 **Proposition 1: DBA cost-neutrality.** Under exact allocation without rounding and without the
73 lower-bound correction, $\sum_h k_h = k_{\text{total}}$. Thus, the total gathered subsequence size is invariant to the
74 entropy distribution.

75 **Implementation.** Entropy statistics are tracked with an exponential moving average:

$$\hat{\mathcal{H}}_h \leftarrow (1 - \mu)\hat{\mathcal{H}}_h + \mu\mathcal{H}_{h,t}.$$

76 Per-head budgets are recomputed periodically to prevent excessive allocation churn while remaining
77 adaptive.

78 3.2 Range Light Cross-Level Guidance

79 **Motivation.** Level- L pooled tokens represent broad context regions. Their selection scores encode
80 which large-scale document regions are likely to be globally important. Conditioning fine-level
81 scoring on this prior can reduce the probability of selecting locally salient but globally irrelevant
82 tokens.

83 **Formulation.** After computing the coarsest-level logits $s_h^{(L)}$, PHAROS upsamples them to the next
84 finer level and injects an additive bias:

$$\tilde{s}_h^{(\ell)} = s_h^{(\ell)} + \gamma_\ell^h \cdot \text{Upsample}_p(\tilde{s}_h^{(\ell+1)}), \quad \ell = L - 1, \dots, 1. \quad (6)$$

85 Here, $\gamma_\ell^h \in \mathbb{R}$ is a per-level, per-head learned scalar initialized to zero, so the mechanism initially
86 recovers the baseline.

87 **Regularization.** To prevent coarse guidance from collapsing fine-level selection into redundant
88 choices, PHAROS adds a KL penalty:

$$\mathcal{L}_{\text{rcg}} = \lambda_{\text{rcg}} \sum_{\ell=1}^{L-1} \text{KL}(\text{softmax}(\tilde{s}_h^{(\ell)}) \parallel \text{softmax}(s_h^{(\ell)})). \quad (7)$$

89 3.3 Tidal Momentum Selection

90 **Motivation.** Transformer models often re-access related tokens across multiple layers [Geva et al.,
91 2022, Yu et al., 2023, Elhage et al., 2021]. Lighthouse Attention computes selection independently at
92 each layer. TMS formalizes the intuition that previous-layer selections provide useful evidence for
93 current-layer selection.

94 **Formulation.** Let $m_h^{(\ell)} \in \mathbb{R}^{T \times P}$ be the tidal state, a running EMA of final selection logits across
95 layers. At layer ℓ :

$$\tilde{s}_h^{(\ell)} = s_h^{(\ell)} + \beta_h \cdot m_h^{(\ell-1)}. \quad (8)$$

96 The tidal state is updated as:

$$m_h^{(\ell)} = \rho \cdot m_h^{(\ell-1)} + (1 - \rho) \cdot \tilde{s}_h^{(\ell)}. \quad (9)$$

97 Here, $\rho \in [0, 1)$ is the momentum coefficient and β_h is a learned per-head scale. The initial tidal state
98 is zero at the start of each forward pass.

Algorithm 1 PHAROS forward pass per transformer layer ℓ and head h

Require: $Q_h, K_h, V_h \in \mathbb{R}^{N \times d}$; tidal state $m_h^{(\ell-1)}$; head budget k_h

Ensure: Output $O_h \in \mathbb{R}^{N \times d}$; updated tidal state $m_h^{(\ell)}$

- 1: Construct pooled pyramids for Q_h, K_h, V_h across levels.
 - 2: Score pooled query-key representations at each level.
 - 3: Inject coarse-to-fine guidance using RCG.
 - 4: Concatenate guided logits across pyramid levels.
 - 5: Add tidal prior from the previous layer using TMS.
 - 6: Update head budgets using DBA.
 - 7: Compute fog mask and select $k_{h,t}^{\text{fog}}$ using FCF.
 - 8: Select top- $k_{h,t}^{\text{fog}}$ tokens and gather $\hat{K}_{h,t}, \hat{V}_{h,t}$.
 - 9: Compute $O_h \leftarrow \text{FlashAttn}(Q_h, \hat{K}_h, \hat{V}_h)$.
 - 10: Update tidal state $m_h^{(\ell)}$.
 - 11: **return** $O_h, m_h^{(\ell)}$
-

99 **Tidal warm-up.** To avoid corrupting early-layer selection with an un-warmed state, PHAROS can
 100 use a layer-indexed gate:

$$\beta_h^{(\ell)} = \beta_h \cdot \sigma(\eta(\ell - \ell_{\text{warm}})). \quad (10)$$

101 3.4 Fog Gate Confidence Fallback

102 **Motivation.** When selector logits are nearly uniform, top- k selection can become unstable. FCF
 103 responds by widening the beam only when the model is uncertain.

104 **Formulation.** Define confidence as:

$$c_{h,t} = \max_j \text{softmax}(s_{h,t})_j. \quad (11)$$

105 A binary fog mask is defined as:

$$\phi_{h,t} = \mathbf{1}[c_{h,t} < \theta_h], \quad (12)$$

106 where θ_h is a learned per-head threshold.

107 When $\phi_{h,t} = 1$, the selection falls back to a wider beam:

$$k_{h,t}^{\text{fog}} = \begin{cases} k_h, & \phi_{h,t} = 0, \\ \min(2k_h, P), & \phi_{h,t} = 1. \end{cases} \quad (13)$$

108 The expected gathered subsequence size is:

$$\mathbb{E}[k^{\text{fog}}] = (1 - f)k_h + f \min(2k_h, P), \quad (14)$$

109 where f is the empirical fog ratio. When the fog ratio remains small, the expected cost stays close to
 110 the baseline.

111 4 Composition and Algorithm

112 The four mechanisms compose in a fixed order that respects their dependencies.

113 **Parameter count.** PHAROS introduces $H(L - 1)$ parameters for RCG, H parameters for TMS,
 114 and H parameters for FCF. Thus, the total additional parameter count is

$$H(L - 1) + H + H,$$

115 which is negligible compared with the size of modern transformer backbones.

5 Theoretical Analysis

Theorem 1: DBA tightens the selection-quality bound. Let $R(k_h)$ denote the expected recall@ k_h of top- k selection relative to the full-attention argmax for head h under a uniform-scores null model. For a non-uniform entropy profile $(\mathcal{H}_h)_{h=1}^H$, DBA reallocates budget from lower-entropy heads to higher-entropy heads while preserving the global budget.

Proof sketch. Heads with lower entropy are in a focused regime where recall saturates early as a function of k . Heads with higher entropy operate in a climbing regime where recall benefits more from additional selected tokens. Under a concave recall curve, transferring budget from saturated heads to unsaturated heads does not reduce total expected recall and can improve it when entropy heterogeneity is present. A complete proof is provided in Appendix A.

Proposition 2: TMS reduces selection variance. TMS smooths selection logits across layers through an EMA recurrence. Under standard assumptions on bounded logit variance and stable momentum coefficients, the smoothed selection signal exhibits reduced layer-to-layer fluctuation relative to independently computed logits. This improved consistency can reduce selection noise during training.

6 Evaluation Protocol

Because unsupported numerical results can misrepresent the method, this submission reports no quantitative claims that have not been independently validated. Instead, the paper defines a reproducible evaluation protocol.

Model and data. PHAROS should be evaluated using the same architecture family, tokenizer, dataset mixture, and training recipe as the Lighthouse Attention baseline. This ensures that any observed differences can be attributed to the selection mechanism rather than unrelated training changes.

Default configuration. The evaluation should compare Lighthouse-style hierarchical attention against PHAROS with DBA, RCG, TMS, and FCF enabled. Hyperparameters should be selected through a validation protocol and reported transparently.

Baselines. Recommended baselines include dense SDPA training or resume baselines, Lighthouse Attention with the norm scorer, Lighthouse Attention with the dilated scorer, and PHAROS with individual mechanisms ablated.

Metrics. Recommended metrics include training loss, resume loss, long-context retrieval accuracy, validation perplexity, throughput, memory usage, and wall-clock latency. All metrics should be reported only after reproducible experimental runs.

7 Results Pending Reproducibility Validation

Quantitative results are intentionally omitted from this draft until independent experiments are completed. This avoids unsupported claims and keeps the paper aligned with responsible reporting standards.

7.1 Ablation Plan

The ablation study should isolate the contribution of each PHAROS component. Each row removes or activates one mechanism while keeping the rest of the training pipeline fixed.

7.2 Fog Gate Analysis Plan

Fog Gate analysis should measure the frequency with which each head activates fallback selection and compare learned thresholds across syntactic, positional, semantic, and global heads. Until head-level statistics are validated, this paper does not report quantitative fog-ratio claims.

Table 1: Planned pretraining comparison. Values should be filled only after reproducible experiments are completed.

System	Train Loss	Resume Loss	Retrieval	PPL	Throughput
Dense SDPA	—	—	—	—	—
Lighthouse, norm scorer	—	—	—	—	—
Lighthouse, dilated scorer	—	—	—	—	—
PHAROS without DBA	—	—	—	—	—
PHAROS without RCG	—	—	—	—	—
PHAROS without TMS	—	—	—	—	—
PHAROS without FCF	—	—	—	—	—
PHAROS, full	—	—	—	—	—

Table 2: Planned ablation study. Values should be filled only after verified experimental runs.

Configuration	Resume Loss	Retrieval	Δ Loss	Δ Retrieval
Lighthouse baseline	—	—	—	—
PHAROS without DBA	—	—	—	—
PHAROS without RCG	—	—	—	—
PHAROS without TMS	—	—	—	—
PHAROS without FCF	—	—	—	—
TMS only	—	—	—	—
DBA only	—	—	—	—
RCG + FCF	—	—	—	—
PHAROS, full	—	—	—	—

159 7.3 Cost and Throughput Analysis Plan

160 The cost analysis should decompose runtime into pooling, scoring, top- k selection, FlashAttention,
 161 and PHAROS-specific overhead. Until verified measurements are available, this draft reports no
 162 wall-clock latency claims.

163 8 Related Work

164 **Sparse attention.** Sparse attention methods such as NSA [Yuan et al., 2025], MoBA [Lu et al.,
 165 2024], HISA [Zhang et al., 2024], and DSA [Chen et al., 2024] select or route attention over subsets
 166 of tokens or blocks. Unlike PHAROS, these methods typically focus on inference-time sparsity
 167 or custom kernels rather than a training-time hierarchical selection wrapper designed to preserve
 168 dense-SDPA compatibility.

169 **Adaptive computation.** Early-exit models [Schuster et al., 2022], Mixture-of-Depths [Raposo
 170 et al., 2024], and CoLT5 [Ainslie et al., 2023] allocate computation adaptively across positions or
 171 layers. DBA applies a related principle orthogonally by redistributing a fixed attention-selection
 172 budget across heads according to entropy.

173 **Cross-layer attention and state.** Recurrent Transformers [Bulatov et al., 2022], Block-Recurrent
 174 Transformers [Hutchins et al., 2022], and Mamba [Gu and Dao, 2023] propagate state across depth or
 175 sequence. TMS is lighter weight: it carries only a scalar-weighted logit EMA, without adding a KV
 176 cache or recurrent hidden state.

177 **Uncertainty in attention.** Bayesian attention [Fan et al., 2021] and Monte Carlo dropout model
 178 uncertainty through probabilistic or sampling-based mechanisms. FCF uses already-computed
 179 selection confidence as a lightweight proxy for uncertainty.

9 Limitations

This work currently presents PHAROS as a method and theoretical framework. The main limitation is the absence of verified quantitative experiments in this draft. As a result, the paper does not claim empirical superiority over Lighthouse Attention or dense SDPA. Future work must validate PHAROS under controlled experimental settings, report statistical variance across runs, and compare memory, latency, and quality metrics against strong baselines.

A second limitation is that the theoretical analysis relies on assumptions about entropy, recall curvature, and cross-layer selection stability. These assumptions should be validated empirically across different model scales, tokenizers, datasets, and context lengths.

10 Conclusion

We presented PHAROS, a four-mechanism augmentation of Lighthouse Attention that addresses structural limitations in hierarchical attention selection. Depth-Sounding Budget Allocation dynamically redistributes token budgets across attention heads based on selection entropy. Range Light Cross-Level Guidance propagates coarse selection priors to fine pyramid levels. Tidal Momentum Selection accumulates selection state across transformer layers. Fog Gate Confidence Fallback widens the attention beam under low-confidence selection.

Together, these mechanisms define an adaptive, cross-level, temporally persistent, and uncertainty-aware framework for hierarchical attention. Future work should validate PHAROS through reproducible long-context experiments, extend the method to vision and audio pyramid architectures, and study whether learned fog thresholds transfer across model scales and tokenizers.

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233 A Proof of Theorem 1

234 Fix head h with budget k_h . Define $\text{recall}@k_h$ as the expected fraction of dense top- k oracle selections
235 recovered by the sparse top- k_h selection:

$$R(k_h) = \mathbb{E} \left[\frac{|\text{top-}k_h(s_h) \cap \text{top-}k_{\text{oracle}}(a_h)|}{k_{\text{oracle}}} \right],$$

236 where a_h is the full-attention score vector and k_{oracle} is fixed.

237 For softmax-distributed scores, recall is assumed to be concave in k , reflecting diminishing returns.
238 High-entropy heads have slower saturation, whereas low-entropy heads saturate earlier. Under
239 uniform allocation, all heads receive the same budget. DBA reallocates budget from lower-entropy
240 heads to higher-entropy heads. By concavity, this transfer does not decrease total expected recall and
241 can strictly improve it when entropy heterogeneity is present.

242 B Fog Gate Head Analysis

243 A complete fog-gate analysis should report the distribution of fallback activation across heads after
244 verified training runs. Until such runs are completed, this appendix does not report quantitative
245 fog-ratio or threshold values.

246 C Complexity Derivation

247 The per-layer complexity of PHAROS beyond Lighthouse-style hierarchical selection is:

- 248 • **RCG**: $O(H \cdot T \cdot P)$ for upsampling and addition per level pair.
- 249 • **TMS**: $O(H \cdot T \cdot P)$ for the EMA update.
- 250 • **DBA**: $O(H \cdot T \cdot P)$ for entropy computation, amortized over update intervals.
- 251 • **FCF**: $O(H \cdot T)$ for per-head, per-position confidence computation.

252 These terms are dominated by the attention computation and selection pipeline, so PHAROS intro-
253 duces no additional asymptotic complexity relative to the hierarchical attention framework.